



Rock Slope Stability Analysis Incorporating the Effects of Intermediate Principal Stress

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Abstract

This paper proposes an analytical approach for assessing rock slope stability based on a three-dimensional (3D) Hoek–Brown (HB) criterion to consider the effects of intermediate principal stress. The 3D HB criterion, considering an associate flow rule, is utilized to describe the perfectly plastic behavior of rock mass under a plane strain condition. To reflect the change of friction angle on the failure surface, the potential failure surface (PFS) is divided into small segments with each segment being assigned a unique friction angle. The upper bound theorem of limit analysis is combined with the strength reduction method to determine the factor of safety (FOS) of a rock slope with a defined PFS. By optimizing the PFS, the minimum FOS and the critical failure surface (CFS) of the rock slope are obtained by the customized genetic algorithm. The proposed approach is validated by comparing it with an HB criterion-based solution and numerical simulations. Parametric studies are also performed to investigate the effects of rock mass properties, slope geometry, and loading conditions on the FOS and CFS. The results indicate that ignoring the 3D strength of rock leads to underestimation of FOS and it is important to consider the various factors when evaluating the stability of a rock slope. For the effortless application of the proposed approach, a Python-based graphical-user-interface application is developed as a stand-alone executable app and is successfully applied to analyze a rock slope.

Highlights

- Three dimensional (3D) Hoek-Brown criterion is applied to analyze rock slope stability
- Customized-genetic algorithm is utilized to accelerate finding most critical condition
- Ignoring 3D strength of rock mass leads to underestimation of the factor of safety for a rock slope
- Graphical-user-interface app was developed using Python for effortless application of the proposed approach

Keywords Slopes · Rocks/rock mechanics · Limit state design/analysis · Factor of safety · 3D strength · Genetic algorithm

List of Symbols

$\vec{n}_i$	Normal unit vector on the failure surface
FOS _{lb} , FOS _{ub}	Lower and upper bounds of FOS
D_{in}	Dissipation rate of internal plastic work
$D_{in,r}$	Dissipation rate of the required stresses
I_1^*, I_2^*, I_3^*	Modified first, second and third stress invariants
$\vec{T}, \vec{X}, \vec{v}$	Traction force vector on boundary; body force vector; velocity vector
W_{ex}	Work rate of external loads
c_1, c_2, c_3	Constants defining the relation between ϕ_{tr} and θ
c_t	Instantaneous cohesion of rock mass
f_{NMZZ}	Failure surface of the newly modified GZZ criterion

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k_v, k_h	Vertical and horizontal seismic coefficients
m_b	Constant parameter of rock mass
m_i	Constant parameter of intact rock
n_{pop}	Population of each generation in the genetic algorithm
q_d	Deviatoric shear stress
r_a	Radius of the failure surface on the top of the slope
r_f	Failure surface determined by ϕ_t
r_{fr}	Failure surface determined by ϕ_{tr}
r_s	Polar function of the slope surface
x_c, z_c	Horizontal and vertical coordinates of the center of rotation
x_t	Horizontal distance from slope toe
γ_{rm}	Unit weight of rock mass
ϵ_{y0}	Initial strain of rock mass in the out-of-plane direction
$\theta_a, \theta_o, \theta_b$	Polar coordinates of the failure surface at points A, O, B (top, corner, and toe)
μ_{rm}	Poisson's ratio of rock mass
ϕ_t	Tangential friction angle of rock mass on the failure surface
$\sigma_1, \sigma_2, \sigma_3$	Major, intermediate, and minor effective principal stresses
$\sigma_1^*, \sigma_2^*, \sigma_3^*$	Modified major, intermediate, and minor effective principal stresses
σ_c	Unconfined compressive strength of intact rock
σ_{ij}	Stress tensor
σ_{ijr}	Required σ_{ij} for stability
σ_n, τ	Normal stress and shear stress
$\sigma_{v,b}$	Vertical stress at the toe of the slope
$\sigma_x, \sigma_y, \sigma_z$	Normal stresses in the horizontal, out-of-plane, and vertical directions
$\sigma_{x0}, \sigma_{y0}, \sigma_{z0}$	Initial values of σ_x, σ_y and σ_z
$\tau_r, c_{tr}, \phi_{tr}$	Required values of τ, c_t and ϕ_t for stability
FOS	Factor of safety
GSI	Geology strength index
D	Disturbance factor representing the effects of construction methods
H	Height of slope
S, V, F	Boundary surface, volume, and failure surface of kinematic failure mechanism
a	Constant parameter of intact rock and rock mass
n	Number of segments constituting the failure surface
q	Load on the top of the slope
s	Constant parameter of intact rock and rock mass
α	Slope angle at the toe of the slope

β	Inclination of the top of the slope
θ	Polar angle measured clockwise from the positive x-direction

1 Introduction

Slope stability assessment is an important research problem in geotechnical engineering. Currently, the linear Mohr–Coulomb (MC) criterion is widely used for analyzing the failure state or critical state of soil and rock in engineering practices such as deep foundations, tunnel excavations, and slope stability. However, the strength of intact rock/rock mass is featured by non-linearity according to laboratory and field tests (Mogi 2007) and thus should be described by a non-linear criterion such as the Hoek–Brown (HB) criterion (Hoek and Brown 1980). Furthermore, the strength of both soil and rock is governed by the three principal stresses under true triaxial stress states, while neither the MC criterion nor the HB criterion considers the intermediate principal stress. Therefore, more recently, many three-dimensional (3D) versions of the MC and HB criteria were proposed by different researchers to reproduce the experimental test results under true triaxial stress states (Zhang and Zhu 2007; Chen et al. 2021a). However, to the best of our knowledge, the application of the 3D MC or HB criterion to slope analysis is still very limited.

Over the past 70 years, different methods have been developed and applied to analyze the stability of soil and rock slopes. The limit equilibrium methods (LEM), especially the methods of slices, were the most widely used in the early years (Bishop 1955; Morgenstern and Price 1965; Spencer 1967). However, the various methods of slices adopt different static equilibrium equations and assumptions about the interactions between slices, and thus they are not rigorous in a true sense and often result in different factor of safety (FOS) values. Later, the upper bound theorem of limit analysis was increasingly applied to evaluate slope stability because no assumptions are needed about interaction forces and predetermined failure surfaces (Lysmer 1970; Chen 1975; Michalowski 1995, 2002). Besides, considering complex geometry and constitutive relations, the limit finite element analysis (LFEA) was also conducted to investigate the slope stability problem (Yu et al. 1998; Loukidis et al. 2003). The displacement finite element method (FEM) and finite difference analysis (FDA), which are more suitable for deformation analyses, were also used for slope stability assessment based on the strength reduction method (Cheng et al. 2007; Tschuchnigg et al. 2015).

Considering the non-linearity of rock strength, attempts have been taken to apply the HB criterion to analyze the rock slope stability in the recent 30 years. Collins et al. (1988) developed an upper bound limit solution to rock slope using the HB criterion. Yang et al. (2004) used the so-called generalized

tangential technique to perform kinematical analyses on rock slopes based on the HB criterion. However, those analyses assume a constant friction angle on the entire failure surface and thus the variation of stress level and friction angle along the failure surface cannot be reflected. To accommodate the change of friction angle and cohesion on a failure surface, Michalowski and Park (2020) divided the failure surface of the upper bound theorem into several segments with each segment corresponding to a specific friction angle and cohesion from the HB criterion. Furthermore, Li et al. (2009) conducted finite element limit analysis (FELA), which uses meshes and considers the changes of stress within the slope, to evaluate the stability of rock slopes based on the HB criterion. Due to the development of commercial numerical software, some design charts or approaches for rock slope stability assessment were also presented (Jiang et al. 2016; Rafei Renani and Martin 2020; Wei et al. 2020) based on commercial software such as FLAC and SLIDE (Rocscience Inc. 2018). However, the accuracy of the numerical model using commercial numerical software highly depends on the mesh quality, and the calculation can be time-consuming.

Optimization methods such as genetic algorithms (Holland 1975) and particle swarm optimization (Kennedy and Eberhart 1995) can iteratively find solutions that can maximize or minimize target function and have been widely used in engineering practices. Genetic algorithms have been applied to analyze rock slope stability by different authors using different analytical or numerical approaches (Li et al. 2010; Gandomi et al. 2017; Zheng et al. 2020; Wijesinghe et al. 2022) and their high efficiency over exhaustive search algorithms has been well proved.

Since the rock slope is under a 3D (true triaxial) stress state, using the HB or MC criteria for slope stability analyses is not suitable due to the ignorance of the intermediate stress. On the other hand, the friction angle of rock mass changes with stress level, and thus the instant friction angle on the failure surface should also change due to the variation of stress. However, none of the available analytical solutions mentioned above can take both the 3D strength and change of friction angle on the failure surface into consideration. Furthermore, numerical rock slope stability analyses can be time-consuming and encounter convergence problems. In this regard, this paper presents an analytical approach to assess the stability of rock slopes based on a 3D HB criterion and the genetic algorithm for optimization. Specifically, to consider the 3D strength, the rock slope is idealized as a perfectly plastic material governed by a 3D HB criterion, the newly modified GZZ criterion (Chen et al. 2021a, b, 2022), with an associate flow rule. To reflect the change of friction angle on the failure surface, the potential failure surface (PFS) is divided into small segments with each segment being assigned a unique friction angle. The upper bound theorem of limit analysis is used to determine the FOS via the strength reduction method for a slope considering a

specific PFS. To ensure accuracy and efficiency, a customized genetic algorithm is utilized to find the minimum FOS and the critical failure surface (CFS) of a slope with specific geometries and loading conditions by optimizing the PFS. With the developed method, parametric studies are performed to investigate the effects of rock mass properties, slope geometry, and loading conditions on the FOS and CFS. Finally, for the effortless application of the proposed method, a Python-based graphical-user-interface (GUI) app is developed as a stand-alone executable program.

2 Newly Modified GZZ Criterion—A 3D Hoek–Brown Criterion

The strength of intact rock/rock mass is featured by non-linearity as indicated by extensive field and laboratory tests (Mogi 2007). Among the different non-linear criteria, the HB criterion is the most widely used in rock mechanics and rock engineering practices, which has the form (Hoek et al. 2002):

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_b \frac{\sigma_3}{\sigma_c} + s \right)^a \quad (1)$$

where

$$m_b = \exp \left(\frac{\text{GSI} - 100}{28 - 14D} \right) m_i \quad (2a)$$

$$s = \exp \left(\frac{\text{GSI} - 100}{9 - 3D} \right) \quad (2b)$$

$$a = 0.5 + \frac{1}{6} \left(e^{-\frac{\text{GSI}}{15}} - e^{-\frac{20}{3}} \right) \quad (2c)$$

in which σ_1 and σ_3 are the major and minor effective principal stresses, respectively; m_i is a constant for intact rock; m_b , a and s are parameters of rock mass; GSI means geology strength index; D is a disturbance factor representing the effects of construction methods.

However, the intermediate principal stress, which is ignored by the HB criterion, has a profound influence on the strength of rock under a 3D stress state (Mogi 2007). In addition, the non-smoothness of the HB criterion at triaxial compression and extension stress states might cause a non-convergence problem in numerical analyses. Therefore, many efforts have been made to develop a 3D and smooth version of the HB criterion. Among them, the newly modified generalized Zhang-Zhu (GZZ) criterion (Chen et al. 2021a, 2021b, 2022) not only reduces to the HB criterion under triaxial stress states but also remains smooth and convex in the 3D principal stress space. Figure 1 compares the HB criterion and the newly modified GZZ criterion in the 3D stress space. In general, compared with the newly modified GZZ criterion, the HB criterion underestimates the strength of rock under a 3D stress state, which is in

the slope. The face collapse and below-toe collapse are not considered as the toe collapse is often the critical case for a homogeneous rock slope. But one cannot exclude the possibility that other failure mechanisms might take place under specific conditions. The rock mass within the failure surface collapses in a rotational mode with every point rotating around point C at a rotational speed of ω . As often assumed (Chen 1975; Michalowski 1995, 2002), the log-spiral failure surface r_f in a polar coordinate system with the center of rotation as the origin in Fig. 2 can be related to the tangential friction angle ϕ_t of the rock mass on the failure surface as

$$r_f = r_a \exp\left(\int_{\theta_a}^{\theta} \tan\phi_t d\theta\right) \quad (8)$$

where r_a is the radius of the failure surface on the top of the slope and denoted by CA in Fig. 2; θ is the polar angle of a point measured clockwise from the positive x -direction in Fig. 2; θ_a is the θ of point A; and ϕ_t is the tangential friction angle of rock mass on failure surface and it changes with stress level.

The ϕ_t is related to the current stress state of rock mass and can be determined by (Chen et al. 2022)

$$\sin\phi_t = \frac{dq_L}{dp_L} = \frac{d\sigma_1 - d\sigma_3}{d\sigma_1 + d\sigma_3} = \frac{d\sigma_1/d\sigma_3 - 1}{d\sigma_1/d\sigma_3 + 1} = \frac{1 - C_{13}}{C_{13} + 1} \quad (9)$$

where

$$C_{13} = -\frac{\frac{\partial f_{\text{NMGZZ}}}{\partial \sigma_1^*} + \mu_{\text{rm}} \frac{\partial f_{\text{NMGZZ}}}{\partial \sigma_2^*}}{\frac{\partial f_{\text{NMGZZ}}}{\partial \sigma_3^*} + \mu_{\text{rm}} \frac{\partial f_{\text{NMGZZ}}}{\partial \sigma_2^*}} \quad (10)$$

And the details of Eq. (10) are given in Appendix 1.

As shown in Fig. 3, the normal stress σ_n and shear stress τ on the failure surface can also be related to the tangential friction angle ϕ_t and instantaneous cohesion c_t as

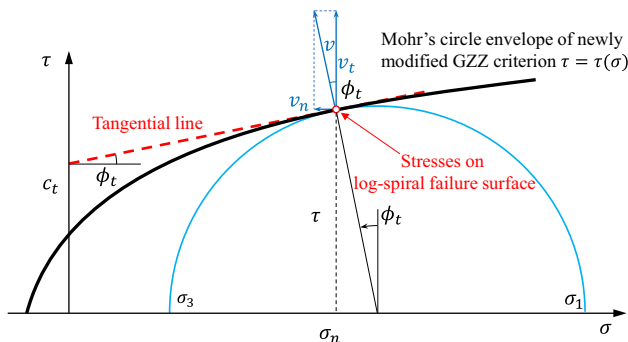


Fig. 3 Stress state on log-spiral failure surface

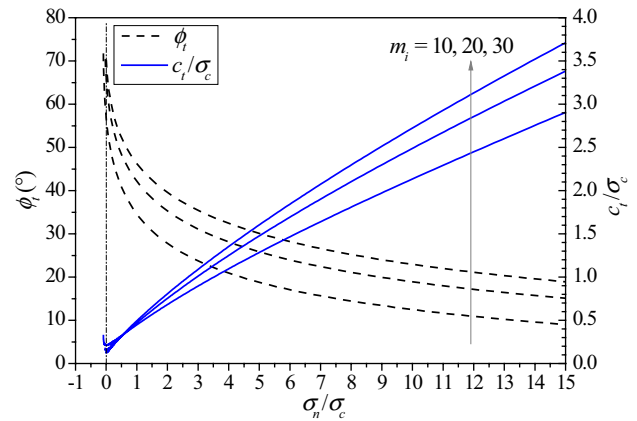


Fig. 4 Relation between ϕ_t , c_t/σ_c and σ_n/σ_c

$$\tau = \frac{\sigma_1 - \sigma_3}{2} \cos\phi_t \quad (11a)$$

$$\sigma_n = \frac{\sigma_1 + \sigma_3}{2} - \frac{\sigma_1 - \sigma_3}{2} \sin\phi_t \quad (11b)$$

$$\tan\phi_t = \frac{d\tau}{d\sigma} \quad (11c)$$

$$c_t = \tau - \sigma_n \tan\phi_t = \frac{\sigma_1 + \sigma_3}{2 \cos\phi_t} - \frac{\sigma_1 - \sigma_3}{2} \tan\phi_t \quad (11d)$$

To better understand the relations among σ_n , ϕ_t and c_t , Fig. 4 shows the σ_n versus ϕ_t and σ_n versus c_t relations of the newly modified GZZ criterion at $m_i = 10, 20, 30$; $D=0$ and $\text{GSI}=100$. As can be seen from the figure, ϕ_t generally decreases with σ_n , while c_t first slightly decreases and then increases with a higher σ_n . At the same σ_n , the ϕ_t increases with a larger m_i , while the c_t decreases when $\sigma_n/\sigma_c < 0.5$ and increases when $\sigma_n/\sigma_c > 0.5$ at a larger m_i . It should be noted that the ϕ_t constituting a PFS would generally decrease with a larger θ as the normal stress on the failure surface would increase with a larger θ . Since the rock mass around the toe should have a smaller vertical stress than a point far behind the slope at the same depth, the σ_3 around the toe would be smaller than the original vertical stress at depth H ($\sigma_3 < \sigma_{v,b} = \gamma_{\text{rm}} H$). By substituting $\sigma_3 = \gamma_{\text{rm}} H$ into Eqs. (1), (7), and (9), the obtained friction angle ϕ_t with $\sigma_3 = \sigma_{v,b}$ should be larger than that at the toe and thus can be regarded as the lower bound of the ϕ_t that constitutes the failure surface according to Fig. 4. It should be noted that if no limitation is set for the friction angle, the friction angle might be underestimated and thus an underpredicted FOS would be obtained by the kinematic limit analysis.

3.2 Upper Bound Limit Analysis Method

The upper bound (kinematic) limit analysis method is utilized here to determine the FOS of the rock slope. According to the theory of classical plasticity (Hill 1948), the criterion governing the material should be convex and an associated flow rule should be incorporated to determine the velocity discontinuity vector in the upper bound approach. In this regard, the newly modified GZZ criterion is convex and thus effortless to be adopted for the rock slope stability problem. The angle between the shear velocity v_t on the failure surface and the velocity on the discontinuity (failure surface) v equals the ϕ_t as shown in Fig. 3. Therefore, the log-spiral failure surface Eq. (8) perfectly satisfies the admissible velocity discontinuity and can be used in the kinematic approach. According to the upper bound limit theorem, the work rate of external loads W_{ex} is not larger than the dissipation rate of internal plastic work D_{in} in any admissible kinematic failure mechanism, i.e.,

$$W_{ex} \leq D_{in} \quad (12a)$$

$$W_{ex} = \int_S \vec{T} \vec{v} dS + \int_V \vec{X} \vec{v} dV \quad (12b)$$

$$D_{in} = \int_V \sigma_{ij} \dot{\epsilon}_{ij}^p dV + \int_F \sigma_{ij} \vec{n}_i \vec{v} dF \quad (12c)$$

in which S , V , and F means the boundary surface, volume, and failure surface (kinematic discontinuity surface) of the kinematic failure mechanism, respectively; $\vec{T}$ is the traction force vector on the boundary surface; $\vec{X}$ is the body force vector (e.g., unit weight and seismic force); $\vec{v}$ is the velocity vector of the kinematic failure mechanism; σ_{ij} is the stress vector of a material point within the kinematic failure mechanism; and $\vec{n}_i$ is the normal unit vector on the boundary surface.

The FOS of the rock slope is defined following the strength reduction method as (Duncan 1996)

$$FOS = \frac{\tau}{\tau_r} = \frac{c_t}{c_{tr}} = \frac{\tan \phi_t}{\tan \phi_{tr}} \quad (13)$$

where τ_r , c_{tr} and ϕ_{tr} are the required values of τ , c_t and ϕ_t for stability. Equation (13) should be satisfied for every point on the failure surface and thus the FOS of each point on the failure surface is the same. According to the limit theorem, the dissipation rate of the required stresses $D_{in,r}$ would be no less than the work rate of the external loads. Therefore, under the critical state,

$$W_{ex} = D_{in,r} = \int_V \sigma_{ijr} \dot{\epsilon}_{ij}^p dV + \int_F \sigma_{ijr} \vec{n}_i \vec{v} dF \quad (14)$$

in which σ_{ijr} is the required σ_{ij} for stability. It should be noted that the ϕ_{tr} now determines the shape and size of the kinematic failure mechanism (i.e., V and F) in Eq. (8) and thus indirectly affects the W_{ex} .

Especially, for the rock slope shown in Fig. 2, the items in W_{ex} , $D_{in,r}$ can be further simplified as

$$\int_S \vec{T} \vec{v} dS = \omega r_0^2 \int_{\theta_a}^{\theta_o} \frac{q \cos \theta \sin^2(\theta_o + \beta)}{\sin^3(\theta + \beta)} d\theta \quad (15a)$$

$$\begin{aligned} \int_V \vec{X} \vec{v} dV = & \gamma_{rm} (1 + k_v) \omega \int_{\theta_0}^{\theta_b} \int_{r_s}^{r_{fr}} r^2 \cos \theta dr d\theta \\ & + \gamma_{rm} k_h \omega \int_{\theta_0}^{\theta_b} \int_{r_s}^{r_{fr}} r^2 \sin \theta dr d\theta \end{aligned} \quad (15b)$$

$$\int_V \sigma_{ijr} \dot{\epsilon}_{ij}^p dV = 0 \quad (15c)$$

$$\int_F \sigma_{ijr} \vec{n}_i \vec{v} dF = \omega \int_{\theta_a}^{\theta_o} r_{fr}^2 (\tau_r - \sigma_n \tan \phi_{tr}) d\theta = \frac{\omega}{FOS} \int_{\theta_a}^{\theta_o} r_{fr}^2 c_t d\theta \quad (15d)$$

$$r_{fr} = r_a \exp \left(\int_{\theta_a}^{\theta} \tan \phi_{tr} d\theta \right) = r_a \exp \left(\int_{\theta_a}^{\theta} \frac{\tan \phi_{tr}}{FOS} d\theta \right) \quad (15e)$$

$$r_s = \begin{cases} r_o \frac{\sin(\theta_o + \beta)}{\sin(\theta + \beta)}; \theta_a \leq \theta < \theta_o \\ r_o \frac{\sin(\theta_o + \alpha)}{\sin(\theta + \alpha)}; \theta_o \leq \theta \leq \theta_b \end{cases} \quad (15f)$$

$$r_o = \sqrt{x_c^2 + z_c^2} \quad (15g)$$

$$\theta_o = 2 \sin^{-1} \sqrt{\frac{r_o + x_o}{2r_o}} \quad (15h)$$

$$\theta_b = 2 \sin^{-1} \sqrt{\frac{H \cot \alpha + x_o + r_b}{2r_b}} \quad (15i)$$

$$r_b = \sqrt{(H \cot \alpha + x_o)^2 + (H + y_o)^2} \quad (15j)$$

where γ_{rm} is the unit weight of rock mass; r_o denotes the length of CO; θ_o and θ_b are the θ values of points O and B;

k_v, k_h are the vertical and horizontal seismic coefficients; r_{fr} is the failure surface determined by ϕ_{tr} in the polar coordinate system; and r_s is the polar function of the slope surface AOB. The detailed expressions for Eq. (15) can be found in Appendix 2.

Equations (14) and (15) are the governing equations for the rock slope problem with the location of the center of rotation (x_c, z_c) and the friction angle and stresses on the failure surface as the basic unknowns. It should be noted that the ϕ_{tr} and c_r change with location as the newly modified GZZ criterion is non-linear. With given x_c, z_c, ϕ_{tr} , the FOS can be solved by substituting Eq. (15) into Eq. (14), which can be numerically solved by a bisection method as detailed in Sect. 3.4.

3.3 Geometric Relations and Constraints of Kinematic Failure Mechanism

To efficiently solve the governing equations Eqs. (14) and (15), the relation and constraints among (x_c, z_c), ϕ_{tr} , τ_r and σ_n should be satisfied. Since the rock mass within the failure surface would all contribute to the work rate of the external forces, the center of rotation (x_c, z_c) should be located above lines BB' and AA' in Fig. 2. Hence,

$$z_c - x_c \tan \beta > 0 \quad (16a)$$

$$z_c - x_c \tan \alpha > 0 \quad (16b)$$

Besides, r_b is larger than r_o given the clockwise rotational mechanism, and thus the rotation center should be above the perpendicular bisector (line DD') of line OB. Therefore

$$z_c + x_c \cot \alpha - H / (2 \sin^2 \alpha) > 0 \quad (17)$$

Equations (16) and (17) define the solution domain for the center of rotation as shown in Fig. 2. And the next step is to find the bounds for the ϕ_{tr} , which defines the shape of the failure surface.

The PFS should be below point O on the slope, which means

$$r_{fr}(\theta_o) = r_a \exp \left(\int_{\theta_a}^{\theta_o} \tan \phi_{tr} d\theta \right) > r_o \quad (18)$$

Combining Eq. (18) with Eq. (15)

$$\exp \left(\int_{\theta_o}^{\theta_b} \tan \phi_{tr} d\theta \right) - \frac{\sin(\theta_o + \alpha)}{\sin(\theta_b + \alpha)} < 0 \quad (19)$$

On the other hand, the PFS is formed from the toe to point A on the slope top. Therefore

$$r_a \sin(\theta_a + \beta) = r_o \sin(\theta_o + \beta) \quad (20a)$$

Substituting Eqs. (15e) and (15f) into Eq. (20a) yields

$$f(\theta_a) = \exp \left[\int_{\theta_a}^{\theta_b} \tan \phi_{tr} d\theta \right] - \frac{\sin(\theta_a + \beta) \sin(\theta_o + \alpha)}{\sin(\theta_o + \beta) \sin(\theta_b + \alpha)} = 0 \quad (20b)$$

By solving Eq. (20b) via a bisection method with a starting range of $[-\beta, \theta_o]$, θ_a can be obtained.

It should be noted that the ϕ_{tr} changes with θ and thus it is better to discretize the failure surface into n segments with each segment being assigned a specific ϕ_{tr} . To ensure the accuracy of calculation, a relatively large $n=50$ is selected based on trial calculations and each segment has the same interval $(\theta_b - \theta_a)/n$. Since the normal stress on the failure surface generally increases from point A to point B (θ increases), the ϕ_{tr} is expected to decrease with a larger θ . It is rational to use an exponential function to relate ϕ_{tr} with θ as

$$\phi_{tr} = c_3 \cdot \exp \left(-\frac{\theta}{c_1} \right) + c_2 \quad (21a)$$

$$0^\circ < c_2, c_3 < 90^\circ \quad (21b)$$

$$c_1 > 0 \quad (21c)$$

where c_1, c_2 and c_3 define the specific relation between ρ_{tr} and θ . Once c_1, c_2 and c_3 are known, the average friction angle ϕ_{tr} on each segment can be determined.

The log-spiral failure surface requires that $\max(\phi_{tr}) < 90^\circ$. Therefore

$$\phi_{tr}(\theta_a) = c_3 \cdot \exp \left(-\frac{\theta_a}{c_1} \right) + c_2 < 90^\circ \quad (22)$$

3.4 Determination of FOS with Given Center of Rotation and Other Parameters

By inputting x_c, z_c, c_1, c_2 and c_3 that satisfy the constraints by Eqs. (16), (17), (19), and (22), the FOS can be determined by a bisection method. As for the bisection method, a continuous governing function needs to be taken as the target function for solving. In the present study, the governing Eq. (14) is revised as the target function with FOS as a variable as

$$F_{FS}(\text{FOS}) = W_{ex}(\text{FOS}) - D_{in,r}(\text{FOS}) = 0 \quad (23)$$

where the elaborate forms of $W_{ex}(\text{FOS})$ and $D_{in,r}(\text{FOS})$ can be found from Eqs. (12), (14) and (15). It should be noted that Eq. (23) is defined to solve Eq. (14) because FOS cannot be explicitly obtained from Eq. (14). And Eq. (23) should be zero, i.e., Eq. (14) is satisfied, if the strength reduction method is well performed (a proper FOS is used). In this

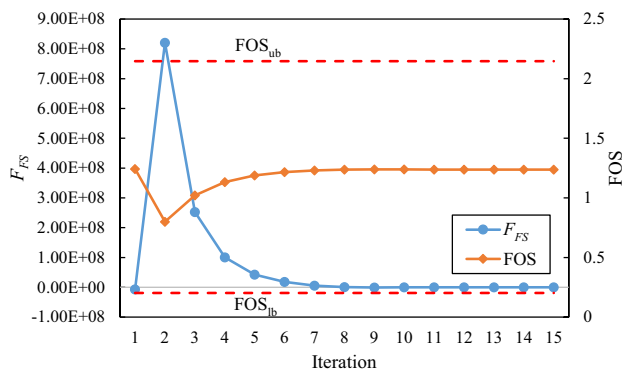


Fig. 5 Evolution of FOS and F_{FS} with iteration

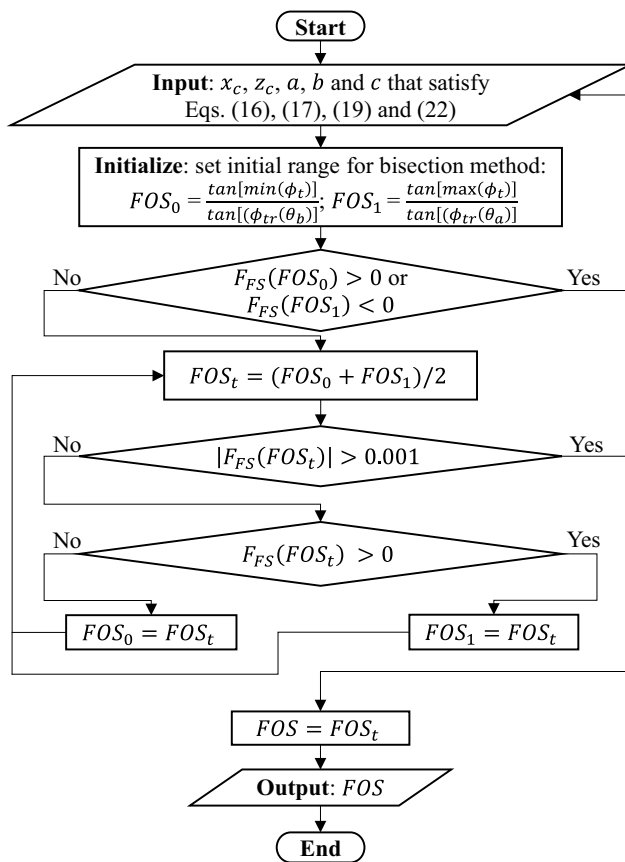


Fig. 6 Flowchart for solving FOS with given x_c, z_c, c_1, c_2 and c_3

case, Eq. (23) is defined as a function of FOS and iterative procedures are performed to solve it and obtain FOS.

As shown in Eq. (15), with given x_c, z_c, c_1, c_2 and c_3 , the work rate of external forces W_{ex} is determined, and FOS only affects the required dissipation rate of internal plastic work

$D_{in,r}$. Generally, $D_{in,r}$ decreases with a higher FOS and thus F_{FS} increases with FOS.

The instant friction angles on the failure surface should fall within $[\min(\phi_t), \max(\phi_t)]$, where $\min(\phi_t)$ is taken as $\phi_t(\sigma_3 = \sigma_{v,b})$ as discussed in Sect. 3.1 and $\max(\phi_t)$ is taken as the friction angle very close to or at the apex of the failure criterion envelope. Therefore, according to the definition of FOS by Eq. (13), the FOS must satisfy

$$\frac{\tan[\min(\phi_t)]}{\tan[(\phi_{tr}(\theta_b))]} = FOS_{lb} < FOS < FOS_{ub} = \frac{\tan[\max(\phi_t)]}{\tan[(\phi_{tr}(\theta_a))]} \quad (24)$$

Therefore, the initial interval (range) of the bisection method would be $[FOS_{lb}, FOS_{ub}]$. The flowchart describing the solving procedures is provided in Fig. 6. It should be noted that other root-searching methods such as Newton–Raphson method (Dedieu 2015) usually requires the derivative of the target function Eq. (23). However, due to the implicit form of Eq. (23), the derivative of it cannot be easily obtained and thus other root-searching methods were not utilized. To better understand the convergence of Eq. (23) using the bisection method, Fig. 5 shows the evolution of both FOS and $F_{FS}(FOS)$ with iteration from one of the case studies performed. As can be seen, the proposed bisection method can converge very quickly and thus is used in this study.

4 Genetic Algorithm-Based Solving Technique

The bisection method presented in Sect. 3 only determines the FOS of a rock slope with pre-defined x_c, z_c, c_1, c_2 and c_3 . However, the true FOS of a slope with specific geometries and loading conditions should be determined by changing the center of rotation and the friction angles on the PFS and finding the most critical condition. In this regard, x_c, z_c, c_1, c_2 and c_3 should be taken as variables and further optimized to obtain the global minimum FOS. In the existing upper bound solutions, the variables were changed one by one in consecutive loops to find the minimum FOS, which might take up to 30 min to calculate the FOS (Michalowski and Park 2020). This process is time-consuming because it is an exhaustive search algorithm and tries to search all possible FOS within the solution domain. Therefore, in this study, a global optimization algorithm, the genetic algorithm, is used to find the minimum FOS. In the following, the genetic algorithm is first briefly described and then a customized genetic algorithm for finding the FOS of a rock slope is presented.

4.1 Brief Introduction of Genetic Algorithm

Genetic algorithm is commonly used to generate high-quality solutions to optimization and search problems in computer science and operations research (Mitchell 1996). It is inspired by the principles of genetics and natural selection and can be used to solve both constrained and unconstrained optimization problems. Generally, the genetic algorithm requires a population of chromosomes and a fitness function (FOS in this study) for optimization. Specifically, a chromosome is basically an array with each element being a variable of the optimization problem, and the fitness function is the objective function used to evaluate a chromosome. With the chromosomes and the fitness function, the genetic algorithm can be performed by initialization of a population of chromosomes within the solution domain and then repetitive applications of the selection, crossover, mutation, and elitism operations, as explained below (Mitchell 1996):

(1) Initialization: An initial population of chromosomes is generated randomly within the solution domain as the first generation. The size of population n_{pop} (numbers of chromosomes) is determined by the number of variables; a larger n_{pop} is used if more variables are involved in the investigated problem.

(2) Selection: Two parents are selected from the current generation with a probability based on the fitness value of each chromosome.

(3) Crossover: A weighted average of the two selected parents to generate an offspring as the new generation, where the weights are random values within $[0, 1]$.

(4) Mutation: Some elements of the generated offspring are mutated by small random changes with small probabilities. Mutation improves genetic diversity and enables the genetic algorithm to search a broader space within the solution domain and thus prevents the algorithm from converging to a local minimum.

(5) Elitism: Crossover and mutation operations are performed repetitively until the size of the generated offspring reaches n_{pop} . Then the offspring and the current generation are merged and sorted in ascending order based on their fitness function values. Only the first n_{pop} chromosomes are saved as the new generation.

The selection, crossover, mutation, and elitism operations are repeated until a termination condition is reached. Common terminating conditions include a maximum number of generations reached and the highest-ranking solution's FOS reaching a plateau.

4.2 Customized Genetic Algorithm for Rock Slope Stability

Following the main procedures of the genetic algorithm, a customized genetic algorithm is developed for evaluating

the stability of a rock slope in this study. The stability assessment of a rock slope is a constrained optimization problem with linear and non-linear constraints as presented in Sect. 3.3. Note that the constraints in Sect. 3.3 cannot be explicitly expressed and thus genetic algorithm packages provided in MATLAB and other computational software cannot be used for solving the current problem. In this regard, a customized genetic algorithm is developed to consider the implicit constraints using Python language. The chromosome is $[x_c, z_c, c_1, c_2, c_3]$ and the fitness function FOS is implicit and can only be numerically solved by a bisection method as provided in Fig. 5. To increase genetic diversity, the population of one generation is set as 50 in the proposed model. The Roulette selection, intermediate crossover, and 3% of mutation rate are specified as detailed in Mitchell (1996). Two terminating conditions are defined: (1) the algorithm stops when the number of generations reaches

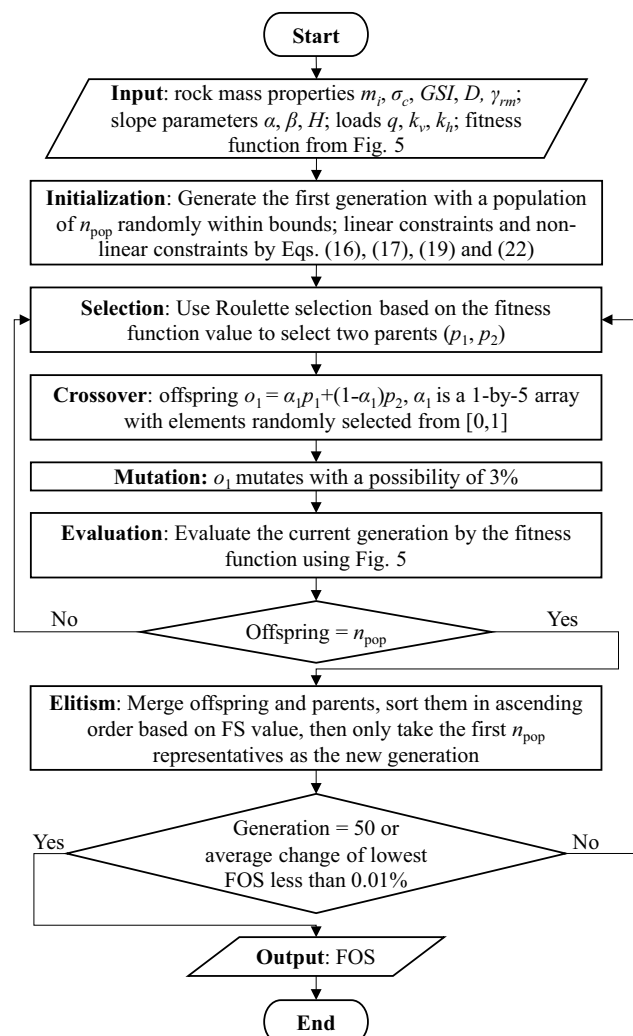


Fig. 7 Flowchart of customized genetic algorithm for determining FOS

Table 1 Comparisons between present and exiting methods

#	α (°)	m_i	D	GSI	$\sigma_c/(\gamma_{rm}H)$	FOS	Equivalent FOS(3D)/FOS(2D)				
							This study	Numerical (2D)		2D HB ^e	This study/ numerical
1	45	10	0	100	1.11	3.623	LEM ^a	3.14	3.152	1.155	1.150
2	45	5	0.7	30	10	1.115	LEM ^a	1.00	0.989	1.115	1.127
3	45	20	0.7	20	8	1.293	LEM ^a	1.20	1.116	1.077	1.158
4	34	7	0.7	16	11.7	1.062	FEM ^b	0.97	0.926	1.095	1.147
5	60	9	1	37	23.8	1.149	FEM ^b	1.14	0.995	1.008	1.155
6	35.5	2	0	5	120	1.632	FEM ^c	1.51	1.411	1.081	1.157
7	45	2	0	5	120	1.352	FLAC ^d	1.15	1.169	1.176	1.157
8	45	5	0	30	40	3.103	FLAC ^d	2.91	2.689	1.066	1.154
9	45	15	0	30	40	3.824	FLAC ^d	3.73	3.305	1.025	1.157
10	45	25	0	30	40	4.293	FLAC ^d	4.19	3.700	1.025	1.160
11	45	15	0	50	40	5.835	FLAC ^d	5.57	5.506	1.048	1.060
12	45	15	0	70	40	10.72	FLAC ^d	10.58	9.333	1.013	1.149
13	30	15	0	30	40	5.478	FLAC ^d	5.20	4.715	1.053	1.162
14	60	15	0	30	40	2.812	FLAC ^d	2.86	2.427	0.983	1.158
15	45	15	0	30	20	3.095	FLAC ^d	3.07	2.666	1.008	1.161
16	45	15	0	30	60	4.388	FLAC ^d	4.32	3.784	1.016	1.160

$$\beta=0^\circ, q=0, k_v=0, k_h=0$$

^aCarranza-Torres (2004)

^bSun et al. (2016)

^cBenz et al. (2008)

^dDetournay et al. (2011)

^eMichalowski and Park (2020)

50, and (2) the algorithm stops when the average change of the FOS of the last ten generations is less than 0.01%. Figure 7 shows the flowchart of the customized genetic algorithm for determining the FOS. The proposed approach was implemented using Python language and it only needs about 2 min to determine the FOS of a slope using a PC with Intel(R) Core(TM) i7-7700 CPU @ 3.60 GHz with default genetic algorithm settings.

5 Comparison and Validation

In this section, the proposed method is compared with some existing methods for evaluating the stability of rock slopes, including LEM, FEM, FLAC simulations, and a kinematic limit solution based on the HB criterion (referred to as 2D HB) in Table 1 (Michalowski and Park 2020). It should be noted that the LEM, FLAC, and FEM models are all based on the 2D HB criterion. To do that, sixteen slopes are selected for comparison, and the rock mass properties and slope geometries of the collected cases are shown in Table 1.

Generally, compared with the existing methods based on the 2D HB criterion, the proposed method gives a higher

FOS because the σ_2 , which enhances the strength of rock, is considered in the adopted newly modified GZZ criterion. Compared with the LEM, FEM, and FLAC simulations, the 2D HB solution predicts a smaller FOS for almost all cases. This is because the 2D HB solution ignores the stress equilibrium of the rock mass and thus too low friction

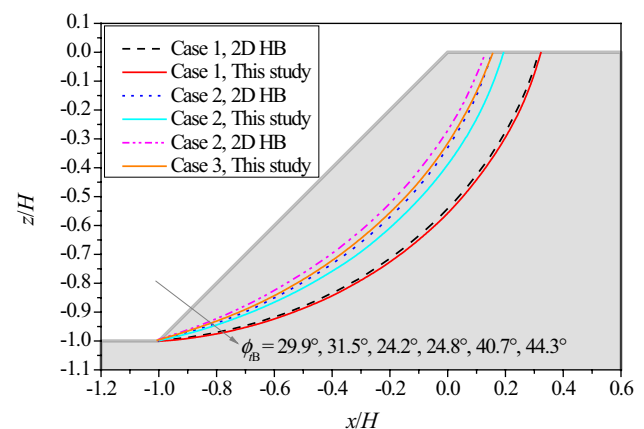


Fig. 8 Effect of 3D strength on failure surface of rock slope

angles might be used in the analyses. In this study, the friction angles are bounded based on the in-situ stress and thus the predicted FOS values are close to but still higher than the LEM, FEM, and FLAC simulations. Therefore, the 3D strength of rock mass and the in-situ stress state can be reasonably considered in the proposed method.

The ratio of the FOS from the present solution to that from the 2D HB criterion-based methods, equivalent $FOS(3D)/FOS(2D)$, is also given in Table 1 to investigate the effect of 3D strength on the FOS. As expected, the equivalent $FOS(3D)/FOS(2D)$ is larger than 1 for almost all cases. Also, the equivalent $FOS(3D)/FOS(2D)$ for the numerical simulations is closer to one than that for the 2D HB-based limit solution. Hence, the proposed method can be used as a tool for fast evaluating the stability of a rock slope.

To further understand the effect of 3D strength on the failure surface, Fig. 8 shows the failure surfaces for cases 1–3 in Table 1. Generally, the 2D HB criterion-based analytical solution gives a lower friction angle at the tow and thus a smaller failure surface than the proposed solution using 3D HB. Therefore, it is important to consider 3D strength because the 2D HB-based solution might underestimate the depth of the failure surface developed into the slope.

6 Parametric Studies

In this section, parametric studies are performed with the proposed method to study the effect of rock mass properties (σ_c , GSI, m_i), slope geometry (H , α , β), and loading conditions (q , k_v , k_h) on the FOS and the CFS (i.e., the PFS with the lowest FOS) of a rock slope. The basic parameters of the rock slope for the analyses are shown in Table 2 and the default values of σ_c , D , H , α , β , q , k_v and k_h are given in parentheses and marked in bold. To be clear, only the results

Table 2 Parameters of rock slope for parametric study

Parameter	Value
m_i	10, 20, 30 (10)
GSI	10–95 (10)
σ_c (MPa)	0.5–100 (10)
D	0, 0.5, and 1.0 (0)
H (m)	10–200 (100)
α (°)	30–90 (45)
β (°)	0–20 (0)
q (kPa)	0–5000 (0)
k_v	0–1 (0)
k_h	0–1 (0)

The default values of m_i , GSI, σ_c , D , H , α , β , q , k_v and k_h for the parametric studies are given in parentheses and marked in bold

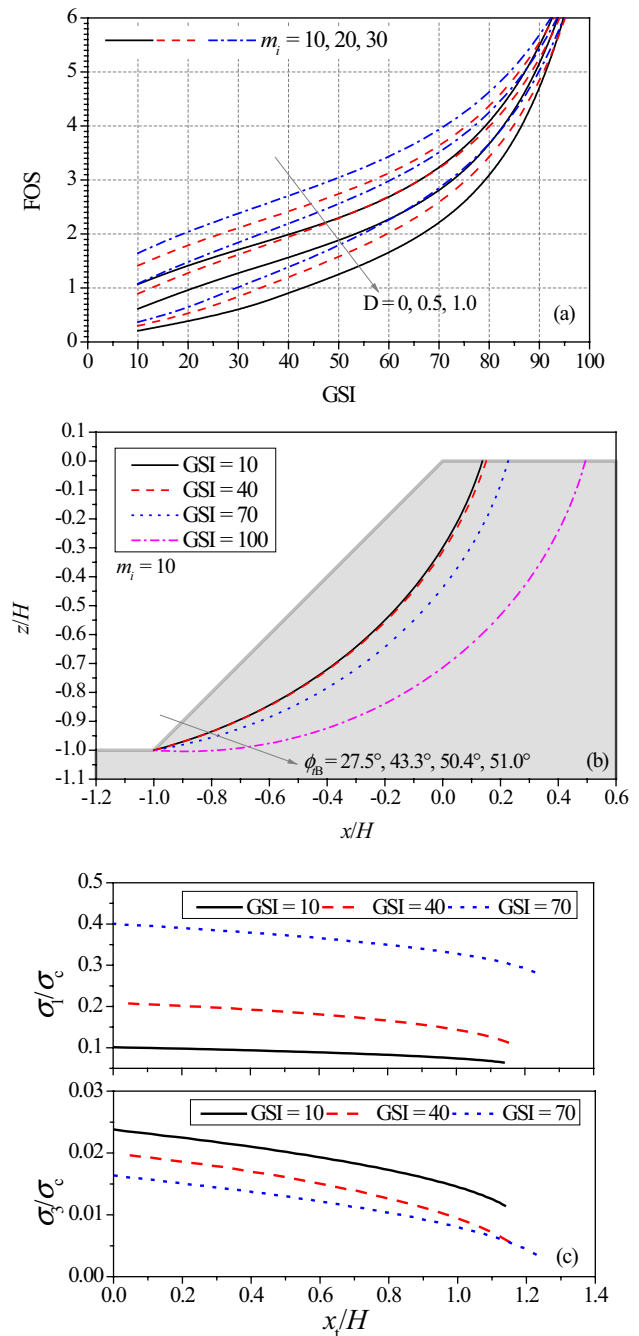


Fig. 9 Effect of GSI on **a** FOS; **b** normalized critical failure surface (CFS) and **c** principal stresses

with $FOS \leq 6.0$ are shown in the figures. Also, if the value of a parameter is not shown in the figure, it means the default value is used.

6.1 Effects of Rock Mass Properties

Figure 9a shows the effect of GSI on the FOS of a rock slope with m_i and D values. Due to the blasting excavation,

the disturbance factor of a rock slope can be larger than zero (Hoek et al. 2002), and thus three D values (0, 0.5, 1) are considered. However, $D=0$ should be assigned to a natural slope. At the different m_i and D , the FOS increases with a higher GSI because the rock mass with a higher GSI has a higher strength. At the same GSI, the FOS increases with a larger m_i but decreases with a higher D . This is again because the strength of rock mass increases with a larger m_i but decreases with a higher D .

Figure 9b investigates the effect of GSI on the normalized CFS of a rock slope with $m_i = 10$. To better understand the differences in the CFS at different GSI, the instant friction angle at the toe of the slope ϕ_{tB} is also shown in the figure. As shown by Eqs. (13) and (15e), the size/shape of the CFS depends mainly on the FOS and ϕ_t on the CFS. When GSI increases from 10 to 100, the FOS increases greatly from 1.1 to 7.9, while the friction angle ϕ_t does not increase as significantly as the FOS (e.g., ϕ_{tB} from 27.5° to 51.0°). In this regard, the $\phi_{tr} = \tan^{-1}\left(\frac{\tan\phi_t}{FOS}\right)$ decreases with a higher GSI and smaller curvatures of the CFS are expected at a higher GSI. Therefore, the CFS expands with a higher GSI due to the smaller curvatures.

Figure 9c shows the variations of σ_1 and σ_3 on the failure surface of a rock slope with $m_i = 10$ and different GSI, where x_t means the horizontal distance measured from the slope toe. As can be seen, both σ_1 and σ_3 decreases with x_t/H (from slope toe to the top surface) at the same GSI, and the curves end at different x_t/H due to the different CFS sizes. And σ_1 increases at a higher GSI due to the higher strength of rock mass. On the other hand, σ_3 decreases at a higher GSI. Note that the variations of σ_1 and σ_3 are related to friction angle and thus will not be discussed in detail for the following analyses.

Figure 10 plots the effect of m_i on the normalized CFS with GSI = 10. The corresponding ϕ_{tB} values are also

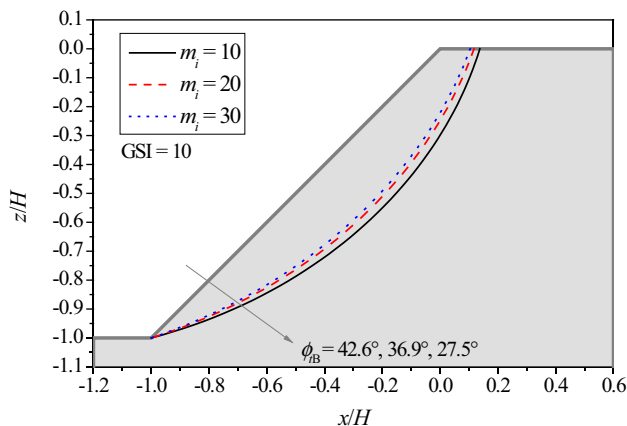


Fig. 10 Effect of m_i on normalized critical failure surface (CFS)

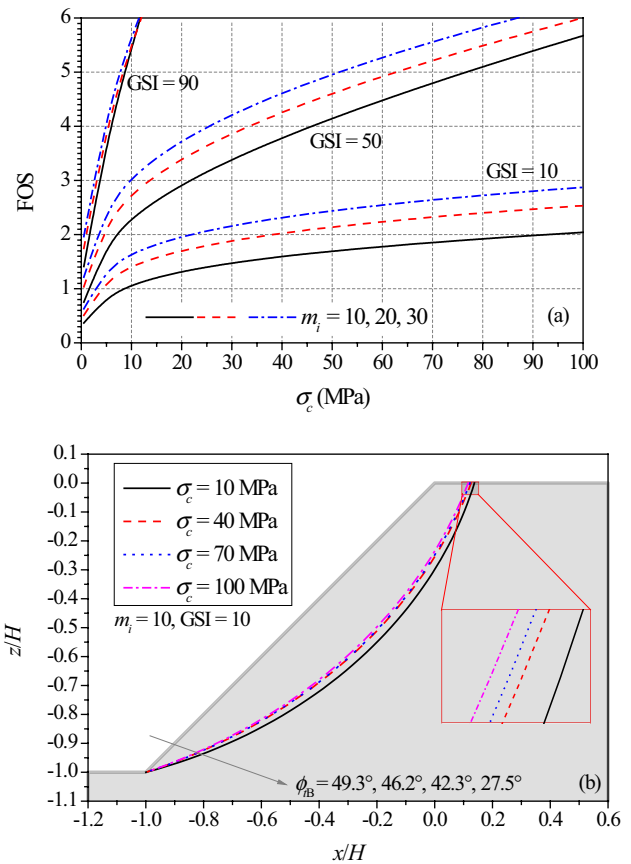


Fig. 11 Effect of σ_c on a FOS; and b normalized failure surface

given in the figure. The CFS shrinks with a larger m_i . This is because when m_i is increased from 10 to 30, the FOS only increases from 1.1 to 1.6 (see Fig. 9a) while the $\tan\phi_{tB}$ increases from 0.52 to 0.92, and thus a larger ϕ_{tr} is expected with a larger m_i based on Eq. (13). As a result, a larger m_i leads to larger curvatures and a smaller size of the CFS.

Figure 11a investigates the effect of σ_c on the FOS of a rock slope with different m_i and GSI values. In this study, the low bound of σ_c is taken as 0.5 MPa because a geo-material with a σ_c less than 0.5 MPa is not regarded as a rock. Generally, the FOS of a rock slope increases with a higher σ_c and the increasing rate decreases with a higher σ_c , simply because a higher σ_c means a higher strength of the rock mass. As expected, the FOS increases with higher m_i and GSI at the same σ_c . It is interesting to note that, at GSI = 90, the FOS barely changes with m_i . And when σ_c is small ($\sigma_c < 1$ MPa), the FOS of the rock slope can be less than 2 even at GSI = 90 and $m_i = 30$, and thus special attention should be paid to rock slopes with a low σ_c (< 5 MPa).

Figure 11b demonstrates the effect of σ_c on the normalized CFS with $m_i = 10$ and GSI = 10. Again, the ϕ_{tB} values at different σ_c are shown in the figure. It is interesting to find

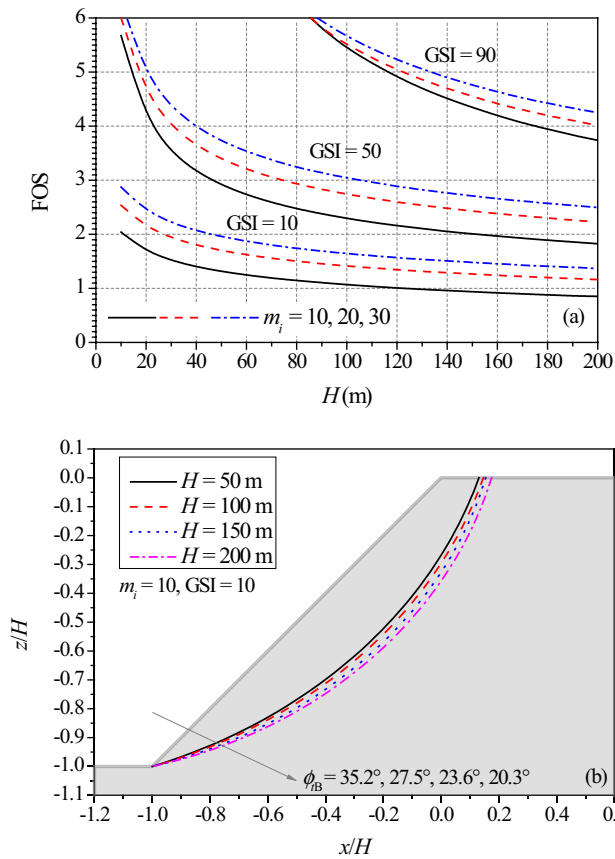


Fig. 12 Effect of H on **a** FOS; and **b** normalized critical failure surface (CFS)

that the ϕ_{tB} increases with a higher σ_c . This is because the stress level at the toe of the slope is barely affected by the σ_c and thus the σ_n/σ_c ratio simply decreases with a larger σ_c . According to the relation between σ_n/σ_c and ϕ_t in Fig. 4, a larger ϕ_t is expected at a larger σ_c due to a smaller σ_n/σ_c . Besides, the FOS doubles (see Fig. 11a) and the $\tan\phi_{tB}$ increases from 0.52 to 1.16 when the σ_c increases from 10 to 100 MPa. This results in the increase of ϕ_{tr} based on Eq. (13) and thus the CFS shrinks with a higher σ_c due to the larger ϕ_{tr} and larger curvatures.

6.2 Effects of Rock Slope Geometry

Figure 12a investigates the effect of slope height H on the FOS of a rock slope with different m_i and GSI values. With the increase of H , the FOS decreases and the decreasing rate becomes smaller. This is because a higher slope imposes larger shear stresses on the CFS. Again, the FOS of a rock slope decreases with a smaller m_i or GSI.

Figure 12b shows the effect of H on the normalized CFS with $m_i = 10$ and $GSI = 10$. The CFS is normalized by H and

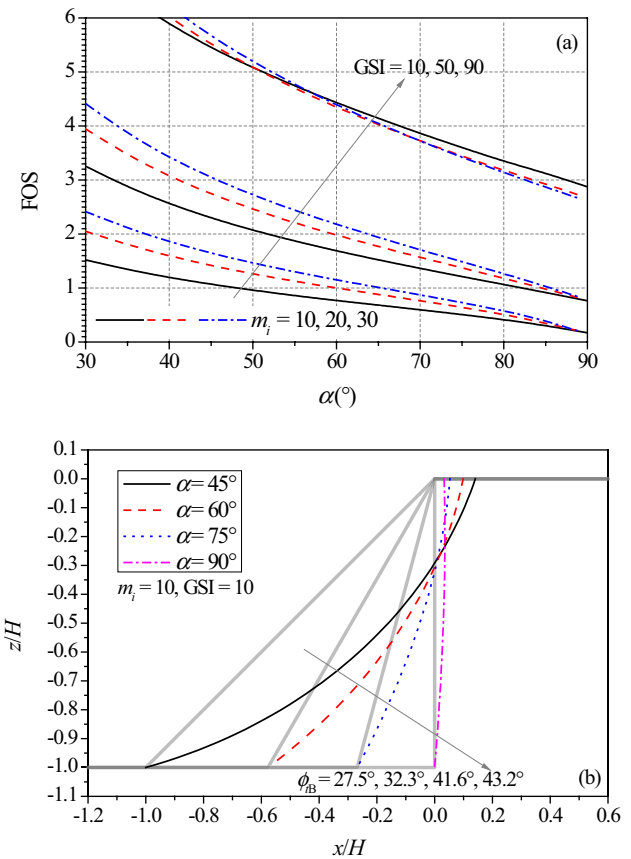


Fig. 13 Effect of α on **a** FOS; and **b** normalized critical failure surface (CFS)

the ϕ_{tB} is shown in the figure. The $\tan\phi_{tB}$ decreases from 0.71 to 0.37 when H increases from 50 m to 200 m due to the higher stress at the toe of a higher slope. Besides, with H increasing from 50 m to 200 m, the FOS decreases from 1.31 to 0.85 (see Fig. 12a), and thus the ϕ_{trB} decreases from 28.2° to 23.5° based on Eq. (13). Therefore, the curvatures of the normalized CFS decrease and the normalized CFS expands with a higher H .

Figure 13a shows the effect of the slope angle at the toe α on the FOS of a rock slope with different m_i and GSI. The FOS almost decreases linearly with a higher α at the different m_i and GSI. Under the same α , the FOS increases with a higher GSI as expected. On the other hand, at the same α , the FOS increases with a higher m_i at a small or medium GSI ($GSI = 10$ or 50), while no consistent trend can be found at $GSI = 90$. This is most likely because the rock mass with a higher m_i might have a smaller instant friction angle when GSI is high ($GSI > 90$) and $\sigma_n/\sigma_c < 0.5$, as shown in Fig. 4. This is in agreement with the results based on the 2D HB criterion (Michalowski and Park 2020).

Figure 13b investigates the effect of α on the normalized CFS with $m_i = 10$ and $GSI = 10$. Again, the ϕ_{tB} is shown in

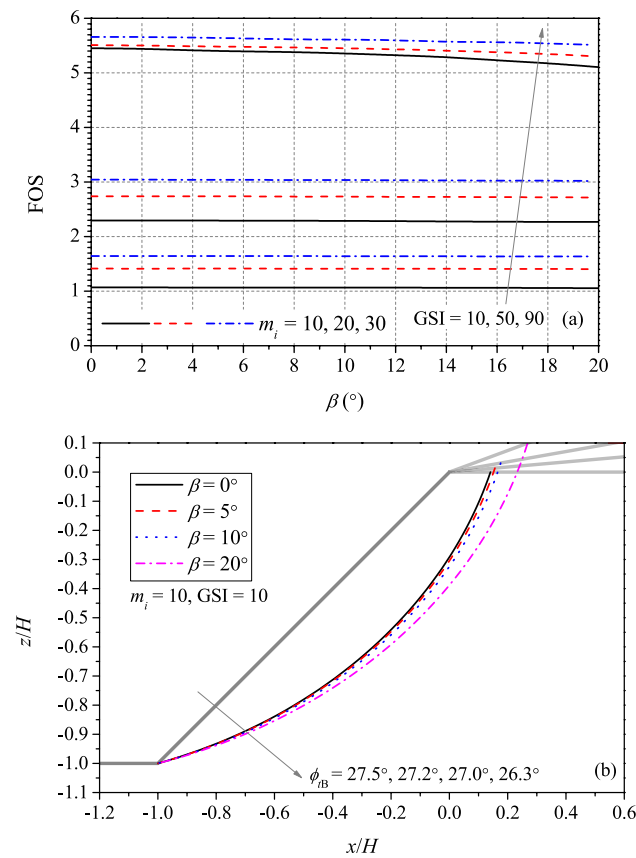


Fig. 14 Effect of β on **a** FOS; and **b** normalized critical failure surface (CFS)

the figure and the CFS is normalized by H . With the increase of α , more significant stress release is expected at the toe of the slope. Therefore, the normal stress σ_n on the CFS at the toe of the rock slope would also decrease with a larger α . In this regard, with the increase of α from 45° to 90°, the ϕ_{tB} is increased from 27.5° to 43.2° and the FOS is decreased from 1.07 to 0.17 (see Fig. 13a). Therefore, ϕ_{tB} increases based on Eq. (13) and thus the curvatures of the CFS increase and the CFS shrinks with a larger α . Although the CFS is significantly small at $\alpha = 90^\circ$, the failure zone with $FOS < 1$ can be much larger because the FOS is only 0.17 on the CFS.

Figure 14a plots the effect of the inclination angle of the slope top face β on the FOS of a rock slope with different m_i and GSI. Compared with α , the β has a much smaller influence on the FOS and the FOS slightly decreases with a larger β , especially at small GSI values. This is because the change of β is smaller than that of α in the parametric study and practices and the β mostly affects the overburden pressure on the top of the slope. As expected, the FOS increases with larger m_i and GSI at the same β .

Figure 14b demonstrates the effect of β on the normalized CFS. A larger β leads to higher overburden pressure around the slope toe and thus a smaller ϕ_{tB} is expected at a

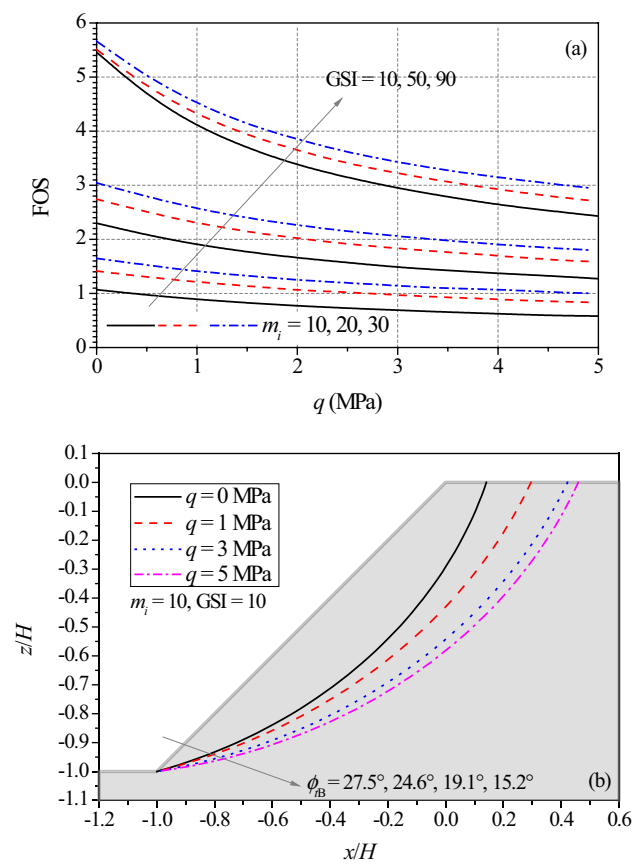


Fig. 15 Effect of q on **a** FOS; and **b** normalized critical failure surface (CFS)

larger β . Under this context, with the increase of β from 0° to 20°, the ϕ_{tB} slightly decreases from 27.5° to 26.3°, the FOS slightly decreases from 1.07 to 1.06 (see Fig. 14a), and thus the ϕ_{tr} also decreases according to Eq. (13). Therefore, the curvatures of the CFS decrease and the CFS expands at a larger β .

6.3 Effects of Loading Conditions

Figure 15a shows the effect of q on the FOS of a rock slope with different m_i and GSI. As expected, the FOS decreases with a higher q at different m_i and GSI, and the decreasing rate increases with a higher GSI but barely changes with m_i . At the same q , a rock slope with larger m_i or GSI has a higher FOS.

Figure 15b investigates the effect of q on the normalized CFS with $m_i = 10$ and $GSI = 10$. With the increase of q from 0 to 5 MPa, the ϕ_{tB} decreases from 27.5° to 15.2° and the FOS also decreases from 1.08 to 0.58 (see Fig. 15a), and thus the ϕ_{tr} decreases based on Eq. (13). Therefore, the curvatures of the CFS decrease and the CFS expands at a higher q .

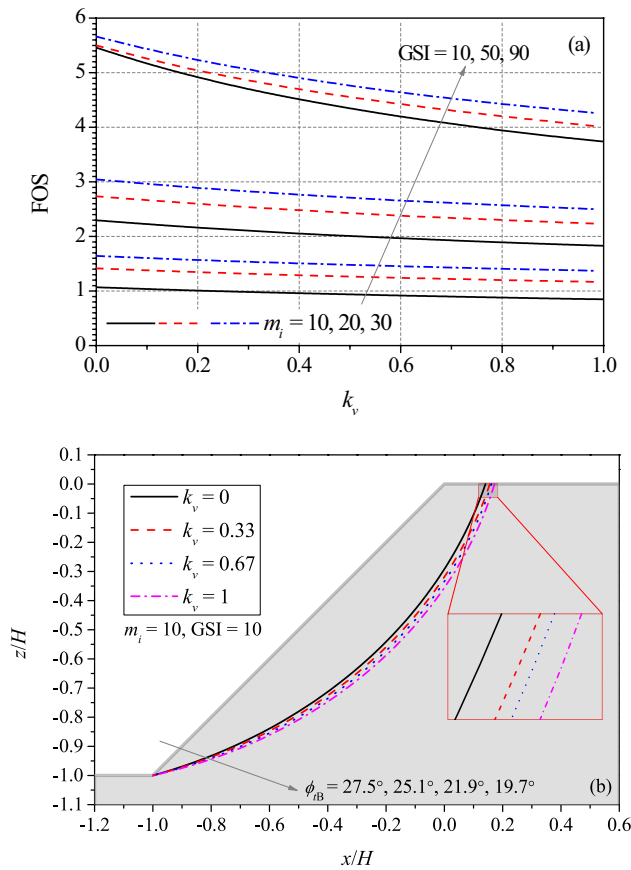


Fig. 16 Effect of k_v on **a** FOS; and **b** normalized critical failure surface (CFS)

Figure 16a demonstrates the effect of k_v on the FOS of a rock slope with different m_i and GSI. Similar to q , the FOS decreases with a higher k_v and the decreasing rate is higher with a higher GSI. This is because the vertical seismic force only imposes vertical body force on the slope, which is equivalent to applying pressure on the top of the slope.

Figure 16b shows the effect of k_v on the normalized CFS with $m_i = 10$ and $GSI = 10$. Similar to the effect of q , the curvatures of the CFS decrease and the CFS expands at a larger k_v .

Figure 17(a) investigates the effect of k_h on the FOS of a rock slope with different m_i and GSI. Compared to the k_v , the k_h has a much greater effect on the FOS; the FOS decreases significantly as the k_h increases and the effect of k_h is more significant at a higher GSI. This is because the horizontal seismic force not only decreases the normal stress but also increases the shear stress on the CFS.

Figure 17b shows the effect of k_h on the normalized CFS with $m_i = 10$ and $GSI = 10$. With the increase of k_h from 0 to 1, the normal stress on the CFS decreases and the ϕ_{tr} increases from 27.5° to 70.1°. Besides, the FOS significantly

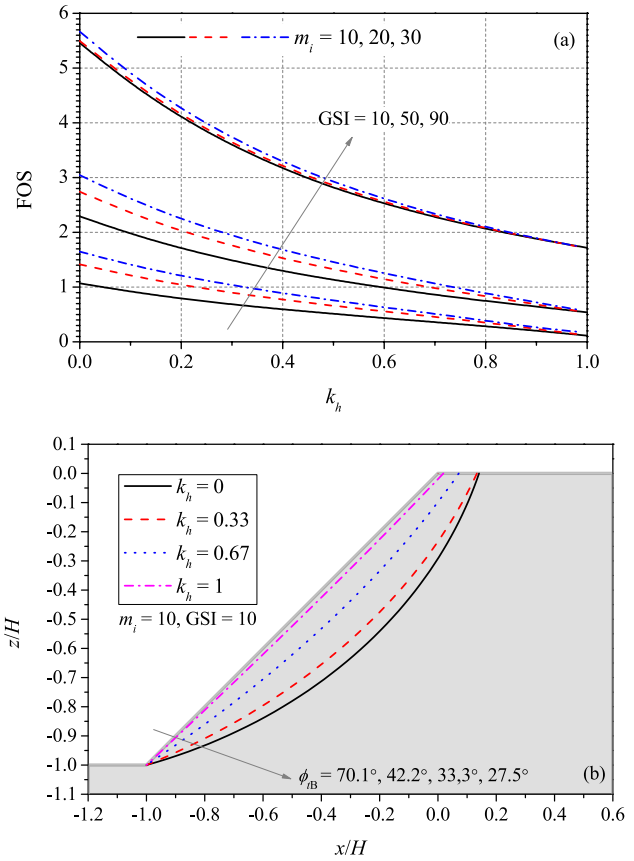


Fig. 17 Effect of k_h on **a** FOS; and **b** normalized critical failure surface (CFS)

decreases from 1.07 to 0.12 (Fig. 17a), and thus the ϕ_{tr} significantly increases when k_h increases from 0 to 1 based on Eq. (13). Therefore, the curvatures of the CFS significantly increase and the CFS shrinks at a larger k_v . It should also be noted that although the CFS is small with $k_h = 1$, the failure zone with $FOS < 1$ can be much larger because the FOS is only 0.12 for the CFS.

7 Application of Proposed Method by a Python GUI App

For convenient application, the proposed genetic algorithm-based method for rock slope stability analysis is implemented in a GUI app based on Python programming language. To develop the GUI app, a Python binding PyQt5 is utilized to create the user interface and a Python package PyInstaller is used to bundle the GUI app and all its dependencies into a single package. The user can run the executable file without installing a Python interpreter. This section would first demonstrate the Python GUI app and then apply it to analyze a rock slope.

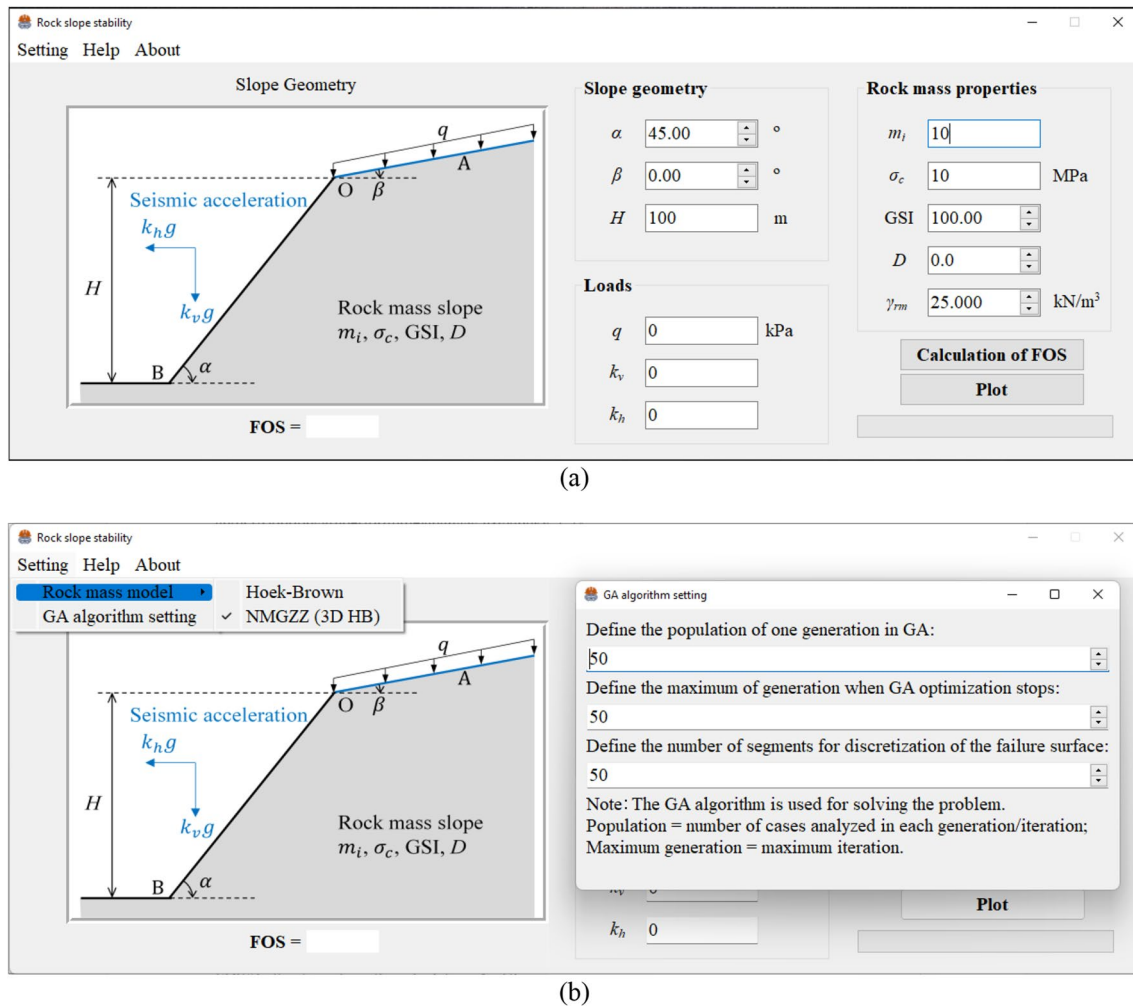


Fig. 18 Python GUI app for using the proposed method: **a** default view; and **b** advanced settings

7.1 Demonstration of Python GUI App

Based on the proposed upper bound limit approach as schemed in Fig. 2, the stability of a rock slope with a height of H and slope angles α and β and under various loading conditions can be assessed. Two major steps are involved in the proposed approach: (1) the assessment of the FOS of a slope with a specific PFS as shown in Fig. 6, and (2) the optimization of the FOS by changing the PFS via a genetic algorithm approach as depicted by Fig. 7. Based on the two flowcharts, a Python GUI application is developed and shown in Fig. 18a. To use it, four steps should be followed to determine the FOS of a slope.

1. Model settings: A user can specify the settings of the rock mass model, the number of failure surface segments, and genetic algorithm options to perform the analysis. By clicking on the "Setting" option on the menu bar, two options are available: "Rock mass model"

- and "GA algorithm setting", as shown in Fig. 18b. For the "Rock mass model" option, one of the two rock mass models, the HB criterion and the NMGZZ (newly modified GZZ) criterion, can be specified. As for the "GA algorithm setting", the user can specify the number of segments of the failure surface during calculation n , the population of chromosomes in each generation n_{pop} and the maximum number of generations n_{max} . The default values of n , n_{pop} and n_{max} are 50, 50 and 50, respectively.
2. Parameters of slope: The user needs to input the rock mass properties (m_i , σ_c , GSI , D and γ_{rm}), slope geometry (H , α and β) and loading conditions (q , k_v and k_h) as shown in Fig. 18a.
3. Run GUI app: Once all parameters are specified, the user can click on the "Calculation of FOS" button to run the calculation. The progress bar below the "Plot" button will show the progress status of the calculation, and the progress bar would turn green when the calculation is finished. The GUI app only needs about 2 min to

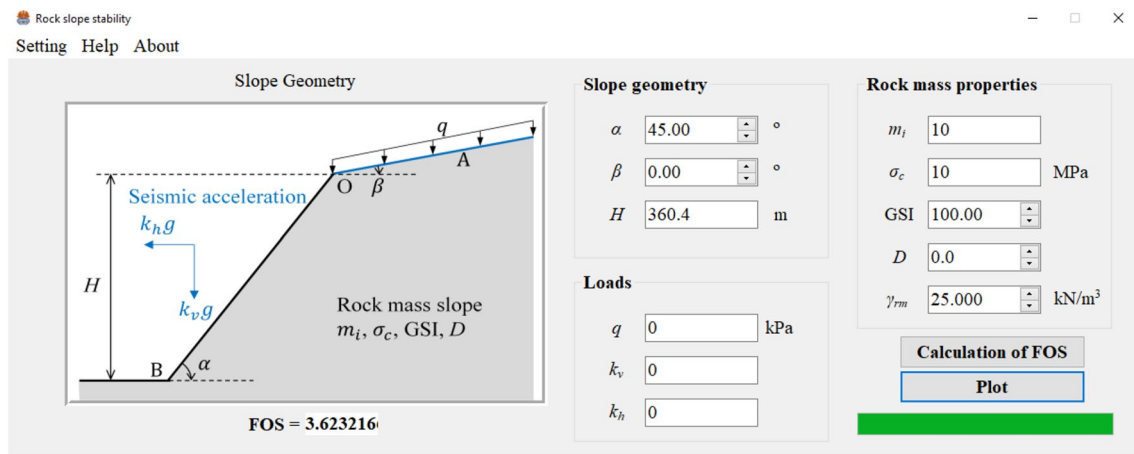


Fig. 19 Application of the Python GUI app to analyze a rock slope

determine the FOS of a slope using a PC with Intel(R) Core(TM) i7-7700 CPU @ 3.60 GHz with default genetic algorithm settings, which is faster than the existing method (Michalowski and Park 2020).

- Results and plots: Once the analysis is complete, the user can use the FOS value below the left figure of the GUI panel. By clicking on the “plot” button, the user can also see figures showing the optimization progress (FOS versus iteration), the CFS and the friction angles on the CFS.

7.2 Application Example

The developed Python GUI app is used to analyze a rock slope from the literature (Carranza-Torres 2004). The rock mass properties, slope geometry, and loading conditions are given in Fig. 19 and Table 1 (#1 case). The σ_c and γ_{rm} of the rock mass are assumed to be 10 MPa and 25000 kN/m³, respectively. By clicking on the “Calculation of FOS” button, the FOS of the rock slope can be obtained as 3.623 as shown in Fig. 17 and Table 1. The calculated FOS value is close to but higher than the result of 3.14 from Carranza-Torres (2004) who used the 2D HB criterion.

8 Conclusion

An analytical approach is proposed for assessing rock slope stability based on a 3D HB criterion and the genetic algorithm. The proposed approach is validated by comparing it with an existing 2D HB criterion-based solution and numerical simulations. Parametric studies using the proposed approach are also conducted to investigate the effects of rock mass properties (m_i , σ_c , GSI), slope geometry (H , α ,

β), and loading conditions (q , k_v , k_h) on the FOS and CFS of a rock slope. The following conclusions can be drawn from the study:

- The proposed approach considers the 3D strength of rock mass and uses the genetic algorithm and thus makes more reasonable and faster predictions on the FOS of rock slopes than the existing methods using an exhaust search algorithm.
- Ignoring the 3D strength of rock mass and the upper bound of the in-situ stress state leads to underestimation of the FOS for a rock slope.
- The rock mass properties (m_i , σ_c , GSI), slope geometry (H , α , β), and loading conditions (q , k_v , k_h) all have great effects on the FOS and the CFS size of a rock slope and should be properly considered when evaluating the stability of a rock slope.
- For the easy use of the proposed approach, a Python-based GUI app was developed. According to the application of the GUI app, it is found that the proposed GUI app provides a simple and easy tool for quick assessment of rock slope stability.

Appendix

Appendix 1

Details of Eq. (10)

$$\frac{\partial f_{\text{NMGZZ}}}{\partial \sigma_i^*} = \frac{\partial f_{\text{NMGZZ}}}{\partial I_1^*} + \frac{\partial f_{\text{NMGZZ}}}{\partial q_d} \frac{3\sigma_i^* - I_1^*}{2q_d} + \frac{\partial f_{\text{NMGZZ}}}{\partial I_3^*} \frac{I_3^*}{\sigma_i^*}; i = 1, 2, 3 \quad (\text{A1a})$$

$$\frac{\partial f_{\text{NMGZZ}}}{\partial I_1^*} = \left[\left(\frac{q_d}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q_d}{\sigma_c} \right)^{1/a-1} \right] \frac{3I_1^{*2} - q_d^2}{3} \quad (\text{A1b})$$

$$\begin{aligned} \frac{\partial f_{\text{NMGZZ}}}{\partial q_d} = & - \left[\left(\frac{q_d}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q_d}{\sigma_c} \right)^{1/a-1} \right] \frac{2q_d I_1^*}{3} \\ & + \left[\frac{2-2a}{a} \left(\frac{q_d}{\sigma_c} \right)^{2/a-2} + m_b \frac{1-a}{a} \left(\frac{q_d}{\sigma_c} \right)^{1/a-1} \right] \left(\frac{I_1^{*3} - I_1^* q_d^2}{3q_d} - \frac{9I_3^*}{q_d} \right) \end{aligned} \quad (\text{A1c})$$

$$\frac{\partial f_{\text{NMGZZ}}}{\partial I_3^*} = -9 \left[\left(\frac{q_d}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q_d}{\sigma_c} \right)^{1/a-1} \right] - 2m_b^2 \quad (\text{A1d})$$

Appendix 2

Details of Eq. (15)

Considering the failure surface in Fig. 2

$$\begin{aligned} \int_{\theta_0}^{\theta_b} \int_{r_s}^{r_{fr}} r^2 \cos \theta dr d\theta &= \int_{\theta_0}^{\theta_b} \frac{r_{fr}^3}{3} \cos \theta d\theta \\ &- \int_{\theta_a}^{\theta_0} \int_0^{r_s} r^2 \cos \theta dr d\theta - \int_{\theta_0}^{\theta_b} \int_0^{r_s} r^2 \cos \theta dr d\theta \end{aligned} \quad (\text{A1})$$

$$\begin{aligned} \int_{\theta_a}^{\theta_0} \int_0^{r_s} r^2 \cos \theta dr d\theta &= \frac{r_o^3}{6} (\tan \theta_o + \tan \beta) \cos^3 \theta_c \\ &\left[\left(\frac{\tan \theta_o + \tan \beta}{\tan \theta_a + \tan \beta} \right)^2 - 1 \right] \end{aligned} \quad (\text{A2})$$

$$\begin{aligned} \int_{\theta_0}^{\theta_b} \int_0^{r_s} r^2 \cos \theta dr d\theta &= \frac{r_o^3}{3} \sin^3 \theta_o (\cot \alpha + \cot \theta_o) \\ &\times \left\{ \left(\frac{\cot \alpha + \cot \theta_o}{\cot \alpha + \cot \theta_b} - 1 \right) (\cot \alpha + \cot \theta_o) - \frac{\cot \alpha}{2} \left[\left(\frac{\cot \alpha + \cot \theta_o}{\cot \alpha + \cot \theta_b} \right)^2 - 1 \right] \right\} \end{aligned} \quad (\text{A3})$$

$$\begin{aligned} \int_{\theta_0}^{\theta_b} \int_{r_s}^{r_{fr}} r^2 \sin \theta dr d\theta &= \int_{\theta_0}^{\theta_b} \frac{r_{fr}^3}{3} \sin \theta d\theta \\ &- \int_{\theta_a}^{\theta_0} \int_0^{r_s} r^2 \sin \theta dr d\theta - \int_{\theta_0}^{\theta_b} \int_0^{r_s} r^2 \sin \theta dr d\theta \end{aligned} \quad (\text{A4})$$

$$\begin{aligned} \int_{\theta_a}^{\theta_0} \int_0^{r_s} r^2 \sin \theta dr d\theta &= \frac{r_o^3}{3} (\tan \theta_o + \tan \beta) \cos^3 \theta_o \\ &\times \left\{ \frac{\tan \beta}{2} \left[1 - \left(\frac{\tan \theta_o + \tan \beta}{\tan \theta_a + \tan \beta} \right)^2 \right] - (\tan \theta_o + \tan \beta) \left(1 - \frac{\tan \theta_o + \tan \beta}{\tan \theta_a + \tan \beta} \right) \right\} \end{aligned} \quad (\text{A5})$$

$$\begin{aligned} \int_{\theta_0}^{\theta_b} \int_0^{r_s} r^2 \sin \theta dr d\theta &= \frac{r_o^3}{6} \sin^3 \theta_o (\cot \theta_o + \cot \alpha) \\ &\left[\left(\frac{\cot \theta_o + \cot \alpha}{\cot \theta_b + \cot \alpha} \right)^2 - 1 \right] \end{aligned} \quad (\text{A6})$$

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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